# Review tables

Counts across the 50 papers in the shared extraction sheet. Paper 45 is left out, since its full text was paywalled and nothing was extracted from it, so every table covers 49 papers and the percentages are of 49. Tables 8, 9, 11 and 13 put each paper in one row only. In the others a paper can appear in several rows, a build that pairs an ESP32 with an Arduino for instance, so those counts come to more than 49.

**Table 1. Hardware platform**

| Feature | Count | % | Papers |
|---|---|---|---|
| ESP32, all variants | 18 | 37 | 2, 12, 14, 17, 20, 23, 26, 28, 29, 31, 32, 42, 43, 44, 47, 48, 49, 50 |
| Arduino (Uno, Nano, Mega) | 18 | 37 | 1, 3, 4, 5, 7, 8, 10, 11, 12, 14, 18, 27, 30, 35, 38, 40, 41, 44 |
| ESP8266 / NodeMCU | 8 | 16 | 7, 15, 20, 35, 38, 46, 49, 50 |
| Raspberry Pi | 8 | 16 | 6, 11, 13, 16, 25, 36, 37, 49 |
| Other or unstated platform | 6 | 12 | 9, 17, 24, 33, 34, 39 |
| Software only, no custom board | 3 | 6 | 19, 21, 22 |

**Table 2. Connectivity**

| Feature | Count | % | Papers |
|---|---|---|---|
| Wi-Fi | 32 | 65 | 3, 6, 7, 10, 11, 12, 13, 14, 15, 16, 17, 20, 21, 22, 23, 25, 26, 28, 31, 32, 33, 35, 38, 40, 42, 43, 44, 46, 47, 48, 49, 50 |
| Named cloud or IoT platform | 17 | 35 | 16, 19, 21, 22, 23, 24, 26, 28, 38, 40, 42, 43, 44, 46, 47, 49, 50 |
| Bluetooth or BLE | 16 | 33 | 1, 4, 6, 10, 11, 16, 21, 22, 25, 30, 33, 42, 44, 47, 48, 49 |
| GSM, cellular or SMS | 11 | 22 | 5, 8, 9, 17, 18, 27, 36, 37, 39, 47, 50 |
| None, or protocol never stated | 3 | 6 | 2, 29, 41 |
| Proprietary RF link to a PC | 1 | 2 | 34 |

**Table 3. Sensors**

| Feature | Count | % | Papers |
|---|---|---|---|
| Camera or image sensor | 12 | 24 | 5, 6, 13, 14, 19, 24, 32, 38, 40, 42, 48, 50 |
| Infrared or proximity | 10 | 20 | 1, 7, 8, 9, 15, 27, 38, 43, 44, 49 |
| Load cell or force sensor | 9 | 18 | 2, 6, 8, 15, 24, 30, 36, 44, 46 |
| Temperature or humidity | 9 | 18 | 3, 8, 12, 14, 25, 32, 37, 38, 47 |
| No sensor, clock or buttons only | 7 | 14 | 11, 18, 23, 29, 31, 35, 41 |
| Reed switch or magnetic switch | 6 | 12 | 7, 14, 17, 26, 33, 34 |
| IMU or accelerometer | 6 | 12 | 1, 21, 33, 34, 42, 49 |
| Vital signs | 6 | 12 | 12, 16, 32, 37, 47, 49 |
| Ultrasonic | 5 | 10 | 1, 10, 13, 36, 37 |
| Fingerprint | 4 | 8 | 17, 36, 37, 38 |
| Microswitch or limit switch | 3 | 6 | 20, 28, 42 |
| Other or unspecified sensing | 3 | 6 | 4, 22, 39 |

**Table 4. User interface**

| Feature | Count | % | Papers |
|---|---|---|---|
| LCD or OLED display | 33 | 67 | 1, 2, 4, 5, 8, 9, 10, 12, 13, 14, 15, 17, 18, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 35, 37, 38, 41, 42, 43, 46, 47, 49, 50 |
| Smartphone application | 31 | 63 | 1, 4, 6, 7, 8, 9, 10, 11, 15, 16, 19, 20, 21, 22, 23, 24, 25, 28, 30, 31, 33, 35, 36, 38, 39, 40, 42, 47, 48, 49, 50 |

| Feature | Count | % | Papers |
|---|---|---|---|
| Buzzer or speaker | 30 | 61 | 3, 5, 8, 9, 10, 12, 15, 17, 18, 20, 25, 27, 28, 29, 30, 31, 32, 35, 36, 38, 39, 40, 41, 42, 43, 44, 46, 47, 49, 50 |
| LEDs | 23 | 47 | 3, 10, 12, 15, 17, 20, 25, 26, 27, 28, 29, 30, 31, 32, 33, 35, 38, 40, 43, 44, 46, 48, 50 |
| Web page or dashboard | 19 | 39 | 2, 10, 12, 13, 14, 15, 16, 17, 19, 22, 26, 31, 32, 38, 40, 43, 44, 47, 49 |
| Buttons, keypad or touch | 18 | 37 | 1, 3, 4, 10, 17, 18, 21, 23, 25, 27, 31, 35, 37, 41, 42, 43, 47, 48 |
| No interface on the device | 1 | 2 | 34 |

**Table 5. Reminder mechanism**

| Feature | Count | % | Papers |
|---|---|---|---|
| Acoustic, buzzer or voice | 32 | 65 | 3, 5, 8, 9, 10, 12, 14, 15, 17, 18, 20, 25, 26, 27, 28, 29, 30, 31, 32, 35, 36, 38, 39, 40, 41, 42, 43, 44, 46, 47, 49, 50 |
| Visual, on LED or screen | 31 | 63 | 1, 2, 3, 4, 10, 12, 14, 15, 17, 18, 20, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 35, 38, 40, 41, 42, 43, 44, 46, 49, 50 |
| Phone or app push notification | 29 | 59 | 1, 2, 5, 6, 7, 8, 9, 10, 11, 15, 16, 19, 20, 22, 23, 24, 28, 30, 31, 35, 36, 38, 40, 42, 43, 44, 47, 49, 50 |
| Escalation after a missed dose | 19 | 39 | 1, 7, 8, 15, 16, 19, 20, 22, 27, 28, 30, 31, 32, 35, 36, 39, 43, 48, 50 |
| SMS, email or telephone call | 9 | 18 | 5, 18, 22, 25, 27, 36, 37, 39, 50 |
| No reminder implemented | 4 | 8 | 13, 21, 33, 34 |

**Table 6. Dispensing mechanism**

| Feature | Count | % | Papers |
|---|---|---|---|
| None, the user takes the pills by hand | 17 | 35 | 11, 12, 17, 18, 19, 21, 22, 23, 25, 26, 28, 29, 30, 33, 34, 36, 47 |
| Servo motor | 16 | 33 | 3, 4, 7, 10, 14, 15, 27, 31, 32, 35, 38, 40, 41, 46, 49, 50 |
| Other or unstated motor | 11 | 22 | 1, 2, 8, 9, 16, 20, 24, 39, 42, 44, 48 |
| Stepper motor | 7 | 14 | 5, 6, 13, 15, 35, 37, 43 |
| Solenoid or electronic lock | 4 | 8 | 6, 27, 35, 40 |

**Table 7. Intake verification**

| Feature | Count | % | Papers |
|---|---|---|---|
| None, or never described | 12 | 24 | 3, 4, 5, 10, 11, 12, 13, 18, 32, 34, 37, 41 |
| Pill removal detected by sensor | 12 | 24 | 8, 9, 15, 16, 20, 27, 38, 39, 42, 43, 44, 49 |
| Self-report, by button or app | 7 | 14 | 23, 25, 28, 31, 35, 47, 50 |
| Weight, force or pill counting | 7 | 14 | 2, 6, 24, 30, 36, 44, 46 |
| Camera or AI image analysis | 6 | 12 | 6, 14, 22, 24, 40, 48 |
| Container access detected | 6 | 12 | 7, 17, 26, 28, 29, 33 |
| Gesture recognition on a wearable | 3 | 6 | 1, 21, 33 |
| QR code scan | 3 | 6 | 19, 42, 48 |

Only papers 1, 14, 21 and 33 try to confirm that the pill reaches the mouth. The rest log that a pill left the device or that the box was opened.

**Table 8. Does the medicine leave its original packaging?**

| Feature | Count | % | Papers |
|---|---|---|---|
| Yes, pills go loose into the device | 36 | 73 | 1, 3, 4, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 17, 23, 25, 26, 27, 28, 29, 30, 31, 32, 34, 35, 37, 38, 39, 40, 41, 42, 43, 44, 47, 49, 50 |
| Not specified | 6 | 12 | 2, 5, 16, 20, 22, 24 |
| No, it stays in its bottle, blister or sachet | 4 | 8 | 33, 36, 46, 48 |
| No container in the system | 3 | 6 | 18, 19, 21 |

Four papers keep the medicine in its packaging, which is what our box is meant to do.

**Table 9. Power supply**

| Feature | Count | % | Papers |
|---|---|---|---|
| Mains, adapter or wired supply | 20 | 41 | 4, 5, 6, 8, 9, 10, 12, 13, 14, 18, 26, 27, 31, 36, 37, 38, 39, 41, 43, 47 |
| Not stated | 16 | 33 | 2, 3, 11, 16, 17, 19, 20, 22, 25, 28, 29, 32, 35, 44, 46, 48 |
| Battery or rechargeable cell | 12 | 24 | 1, 7, 21, 23, 24, 30, 33, 34, 40, 42, 49, 50 |
| Mains with battery backup | 1 | 2 | 15 |

**Table 10. Caregiver support**

| Feature | Count | % | Papers |
|---|---|---|---|
| Notifications or alerts | 33 | 67 | 2, 3, 6, 7, 8, 10, 11, 13, 14, 15, 16, 17, 19, 20, 22, 24, 25, 27, 28, 31, 32, 36, 37, 38, 39, 40, 42, 43, 44, 47, 48, 49, 50 |
| Remote monitoring of adherence | 28 | 57 | 2, 4, 7, 8, 9, 10, 11, 13, 15, 16, 17, 18, 19, 20, 22, 28, 31, 32, 33, 35, 38, 39, 44, 46, 47, 48, 49, 50 |
| Clinician or pharmacist dashboard | 16 | 33 | 16, 17, 19, 22, 31, 32, 33, 35, 36, 37, 38, 39, 44, 47, 48, 49 |
| Remote schedule or therapy setup | 12 | 24 | 2, 7, 9, 10, 12, 18, 28, 31, 35, 43, 49, 50 |
| None, or not specified | 9 | 18 | 1, 5, 21, 23, 26, 29, 30, 34, 41 |

**Table 11. Maturity of the work**

| Feature | Count | % | Papers |
|---|---|---|---|
| Tested, working prototype | 33 | 67 | 1, 2, 5, 6, 7, 9, 10, 12, 13, 14, 16, 17, 18, 20, 21, 23, 25, 26, 28, 30, 33, 34, 35, 36, 37, 39, 41, 42, 43, 44, 46, 49, 50 |
| Preliminary prototype | 11 | 22 | 3, 4, 8, 11, 27, 29, 31, 38, 40, 47, 48 |
| Design or framework only | 5 | 10 | 15, 19, 22, 24, 32 |

**Table 12. Evaluation carried out**

| Feature | Count | % | Papers |
|---|---|---|---|
| Bench or functional test | 31 | 63 | 1, 2, 5, 6, 7, 9, 10, 12, 13, 14, 17, 18, 20, 23, 25, 26, 27, 28, 30, 31, 33, 37, 38, 41, 42, 43, 46, 47, 48, 49, 50 |
| No test reported | 8 | 16 | 3, 4, 8, 15, 19, 24, 29, 40 |
| Usability or lab study | 8 | 16 | 12, 21, 25, 31, 33, 34, 41, 42 |
| Field or in-home trial | 7 | 14 | 20, 35, 36, 39, 42, 44, 46 |
| Simulation only | 6 | 12 | 11, 16, 31, 32, 38, 44 |
| Validation on a public dataset | 2 | 4 | 22, 36 |

**Table 13. Relevance to our project**

| Feature | Count | % | Papers |
|---|---|---|---|
| High | 20 | 41 | 7, 10, 13, 14, 16, 17, 22, 26, 28, 30, 33, 35, 36, 40, 42, 43, 44, 46, 49, 50 |
| Medium | 20 | 41 | 1, 5, 6, 8, 9, 12, 15, 20, 21, 23, 24, 27, 29, 31, 34, 38, 39, 41, 47, 48 |
| Low | 9 | 18 | 2, 3, 4, 11, 18, 19, 25, 32, 37 |